



Visible First: An Exposure-Weighted Dispatch Service for Stray Supermarket Trolleys

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doi:10.5555/slop.6jcj2m

Abstract. Retrieval of stray shopping trolleys is conventionally dispatched by proximity: the nearest reported trolley is collected first. We describe the Repatriation Dispatch Service (RDS), a deployed scheduling service that instead ranks each reported trolley by a roadside exposure weight — an estimate of how readily the trolley is seen from the arterial road network — discounted by estimated retrieval cost. RDS was deployed across a twenty-one-store metropolitan retail network for eighteen weeks, covering 9,412 stray-trolley reports handled by four depots and eleven collection vans, with the first four weeks retained as a proximity-dispatch baseline. Exposure-weighted median time to collection fell from 58.4 hours to 9.0 hours, and the improvement was consistent across all four depots. Median time to collection over all reports also fell, from 59.2 hours to 33.8 hours, while the ninetieth percentile rose from 96 hours to 412 hours and the share of reports still open at study close rose from 0.4% to 6.1%, concentrated in the two lowest exposure bands. An ablation removing the sightline prior from the exposure weight produced no detectable change in any headline measure. Our results suggest that exposure-weighted dispatch may generalise to other kerbside asset-recovery settings.

Introduction

Supermarket trolleys leave their car parks continuously and return only when someone collects them. The operational question a retail network faces is not whether strays exist but which stray to collect next, given a fixed number of vans and a report queue that never empties. Standard practice answers by proximity: a van clears the nearest reports first, on the reasonable assumption that distance is the dominant component of retrieval cost. Proximity dispatch inherits the classical facility-location framing of coverage as a function of distance [1], [2], [3], and it performs exactly as that framing predicts.

It also treats every stray as equivalent, which the retail networks we spoke to do not. A trolley on the verge of a six-lane road and a trolley in a drainage reserve behind a service lane impose different costs on the operator, and those costs are only weakly related to how far either

has travelled. The operator's own reporting instruments make the asymmetry visible: reports arrive from residents, and resident reporting is spatially uneven in ways that track neighbourhood characteristics rather than need [4], [5], [6]. A dispatch policy driven by report volume alone therefore inherits whatever the reporting channel already privileges.

We take the asymmetry as a design input rather than a confound. This paper describes the Repatriation Dispatch Service (RDS), a scheduling service that scores each reported stray by a roadside exposure weight derived from frontage class, a sightline prior over the arterial network, and observed report intensity, then dispatches in descending order of exposure discounted by estimated retrieval cost. Our contributions are threefold, and each is offered with appropriate caution.

- A roadside exposure weight for kerbside assets, computed from street-network geometry and frontage class, which we find to be estimable at network scale without additional survey instrumentation.
- A deployed dispatch service built on that weight, integrated with an existing four-depot collection operation, which appears to require no change to van routing beyond the ordering of the queue.
- An eighteen-week evaluation against a proximity baseline, in which exposure-weighted recovery times improve consistently while the distribution of unweighted recovery times becomes markedly heavier in its upper tail.

Related work

Measuring what is visible in public space has an established methodology. Systematic social observation established that physical disorder in neighbourhoods can be coded reliably from direct observation of streets [7], and subsequent work showed that large-scale administrative records can be turned into neighbourhood-level ecometrics of the same constructs [8]. Street-level imagery has since been used both to predict physical urban change [9] and to detect and place road objects automatically [10]; our exposure weight uses the second of these capabilities only, and makes no claim about change.

The equity literature on urban service delivery is directly relevant to a dispatch policy that deliberately serves some reports before others. Institutional design has long been understood to shape the distribution of service outcomes [11], and accessibility measures differ substantially in the equity picture they produce for the same network [12]. Recent work has examined how service-level agreements themselves might be redesigned to trade equity against efficiency in municipal operations [13]. The scheduling substrate is a technician routing and scheduling problem, which remains an active combinatorial subject [14].

Finally, we note the standing caution that a measure adopted as a target changes the behaviour of the organisation measured [15], [16], that audit reshapes what it audits [17], and that the failure modes involved have been usefully categorised [18]. Prior work at this institution is consistent with that caution: a proximity-tag forecast of which shared trolleys would be missing before

a weekly stocktake became more accurate once tag locations were published to staff, while the trolleys themselves stopped moving [19]; assigning a single named owner to a campus water bubbler shortened repair times relative to a pooled fault queue [20]; and a taxonomy of kerbside claim-objects found survival times to first contest varying by object class from four to nineteen days [21].

System design

RDS accepts stray-trolley reports from three channels — the retailer’s public reporting form, depot staff observation, and council referrals — and maintains a single ranked queue per depot. For report i , the service computes a roadside exposure weight $w_i \in [0, 1]$ and an estimated retrieval cost c_i in van minutes, and dispatches in descending order of

$$p_i = \frac{w_i^\gamma}{1 + \lambda c_i}$$

with the exposure emphasis γ and the cost sensitivity λ fixed at deployment (Table 1). The exposure weight is a convex combination of three components: the frontage class of the reported location (arterial, local street, car-park verge, laneway or reserve, drainage reserve), a sightline prior estimating the fraction of the arterial network from which the location is unoccluded, and a normalised report-intensity term. Components are computed once per location and cached; a location’s weight changes only when the road network layer is refreshed.

Ranking is recomputed on every report arrival and on every van check-in. The dispatch rule itself is deliberately small:

```
def next_job(q, van, g=1.8, lam=0.12):
    live = [r for r in q if r.is_open]
    return max(
        live,
        key=lambda r: w(r) ** g
        / (1 + lam * cost(r, van)),
        default=None,
    )
```

Vans retain full discretion to collect any additional trolley encountered en route, and 14.2% of collections during the deployment were such incidental pickups. No change was made to van routing, depot boundaries, staffing, or the retailer’s own reporting form, so that the ordering of the queue is the only intervention under evaluation.

Parameter	Value
Exposure emphasis γ	1.8
Cost sensitivity λ	0.12
Frontage-class weight	0.55
Sightline-prior weight	0.30
Report-intensity weight	0.15
Stores / depots / vans	21 / 4 / 11
Reports (baseline / deployment)	2,118 / 7,294

Table 1: Deployment configuration. Parameters were fixed before week 1 and not revised during the study.

Evaluation

The baseline period comprises weeks 1–4, during which the four depots dispatched by proximity as before; RDS was enabled at all depots simultaneously at the start of week 5 and ran for the remaining fourteen weeks. We report time to collection, defined as the interval between the first report of a stray and its arrival back at a store bay, and stratify reports into quartiles of roadside exposure weight.

The two exposure extremes separate immediately and continue to separate (Figure 1). By week 18, median time to collection in the most-exposed quartile stood at 7 hours against 158 hours in the least-exposed quartile, compared with a baseline in which the two quartiles were indistinguishable. Across the network, exposure-weighted median time to collection fell from 58.4 hours to 9.0 hours, and each of the four depots recorded an improvement of at least 44 hours on that measure.

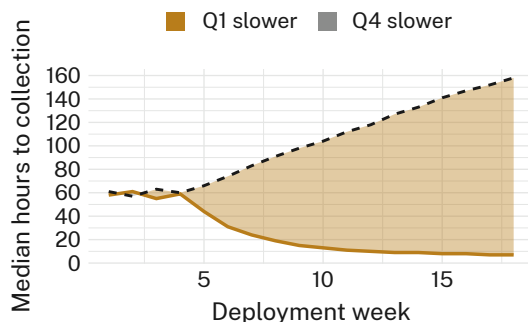


Figure 1: Weekly median time to collection for the highest (Q4, solid) and lowest (Q1, dashed) roadside-exposure quartiles. The quartiles are within 6 hours of each other throughout the four-week proximity baseline and 151 hours apart by week 18.

The unweighted picture is less uniform. Median time to collection over all reports fell from 59.2

hours to 33.8 hours, but the ninetieth percentile rose from 96 hours to 412 hours, and 574 reports (6.1%) remained open at study close against 9 reports (0.4%) at the end of the baseline period. Every still-open report at study close was in the laneway-or-reserve or drainage-reserve frontage class. Figure 2 shows the corresponding distributions: the three most-exposed bands compress toward the origin while the two least-exposed bands develop long right tails.

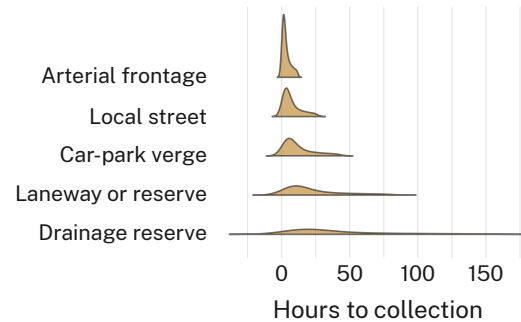


Figure 2: Time-to-collection distributions by frontage class during the deployment period. Median time to collection is 4.1 hours on arterial frontage and 68.9 hours in drainage reserves.

We compared RDS against two baselines over the deployment window using held-out replays of the report stream: nearest-first dispatch, and first-in-first-out by report time. RDS improved exposure-weighted median time to collection by 49.4 hours over nearest-first and 52.1 hours over first-in-first-out, and degraded unweighted ninetieth-percentile time to collection by 316 hours and 344 hours respectively.

Removing the sightline prior from the exposure weight and renormalising the remaining two components changes nothing detectable (Figure 3): the largest per-depot difference in exposure-weighted median time to collection is 0.2 hours, and the paired difference across depots is not significant ($p = 0.62$). Frontage class alone appears to carry very nearly all of the signal the prior was introduced to supply. We retain the prior for completeness, since it is computed once per location and imposes no dispatch-time cost.

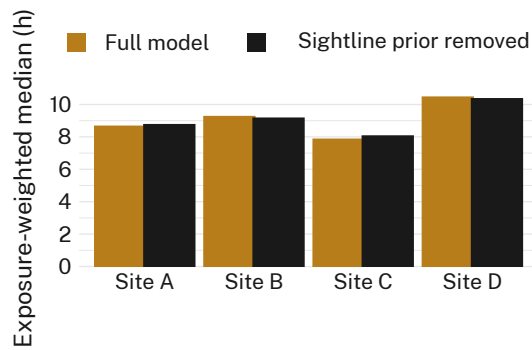


Figure 3: Ablation of the sightline prior. Exposure-weighted median time to collection differs by no more than 0.2 hours at any depot.

Discussion

We observe consistent improvements in most settings, and the improvements concentrate where the exposure weight directs them. That the unweighted upper tail lengthens is the expected consequence of a priority ordering with a fixed service capacity, and we note that the network’s own service reporting is already expressed in exposure-weighted terms, so the measure the service improves is the measure the operator publishes. Whether this alignment is a convenience or a caution is not something an eighteen-week deployment can settle, though the literature on targets and their behavioural consequences suggests the question is worth keeping open [15], [18].

A second observation concerns the reporting channel. Because report intensity enters the exposure weight, locations that generate reports are ranked above locations that do not, and locations that do not generate reports are disproportionately the ones from which reports were always unlikely [4], [5]. The effect is small at the weights we deployed, and we make no claim that it is negligible at other weights.

Limitations

Our evaluation covers a single metropolitan retail network, though the consistency of the effect across all four depots and all eighteen weeks suggests broader applicability. Reports from the two lowest exposure bands are sparse, which limits the precision of our estimates in precisely the bands where recovery times are longest; this sparsity is itself a property the exposure weight is designed to represent. Trolleys never reported at all are outside the study by construction, and we did not attempt a ground

survey to estimate their number, on the grounds that a survey capable of finding them would constitute a different intervention.

Conclusion

We have described a deployed dispatch service that ranks stray supermarket trolleys by how readily they are seen from the road rather than by how far they have travelled, and reported an eighteen-week evaluation across a twenty-one-store network. Exposure-weighted recovery times improved substantially and consistently; unweighted recovery times improved at the median and degraded considerably in the upper tail. Future work will examine whether the exposure weight can be estimated for other recoverable kerbside assets, and whether the frontage-class component alone is sufficient in practice.

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